

Spectrum

Spectrum

- Sources of spectrum: Bunsen-carbon-mercury-sodium Emission and absorption spectra
Classification of emission spectra Solar spectrum
Ultraviolet Spectrum Infra red spectrum
Electromagnetic spectrum

Introduction - spectrum

- **Spectrum**, in optics, the arrangement according to wavelength of visible, ultraviolet, and infrared light
- An instrument designed for visual observation of spectra is called a spectroscope; an instrument that photographs or maps spectra is a spectrograph
- In Latin, *spectrum* means "image" including the meaning "spectre".
- Spectroscopy is the study of objects based on the spectrum of color they emit, absorb or reflect.

Definition of White Light

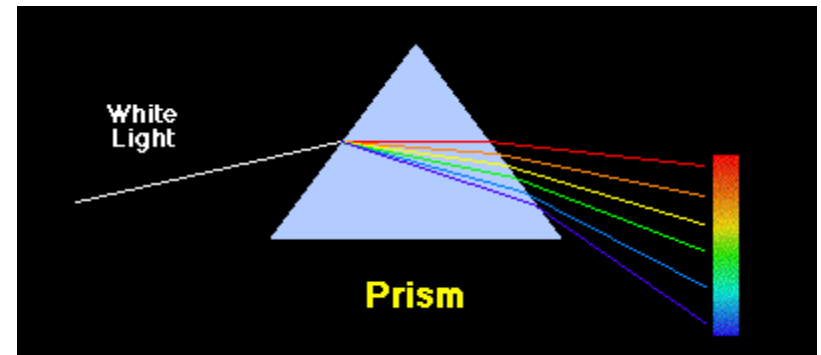
- **White light** is defined as the complete mixture of all of the wavelengths of the visible spectrum.
- This means if the beams of light of all of the colors of the rainbow are focused onto a single spot, the combination of all of the colors will result in a beam of white light.

Sources of White Light

- White light can be generated by a variety of sources both in space and by artificial sources on earth.
- For example, the sun and other stars are sources of white light.
- The sun is the most obvious source of white light in our solar system.
- As for artificial sources, fluorescent light bulbs and white LEDs produce white light.
- Other light bulbs, like the incandescent lamp, do not produce white light.
- They produce light of much longer wavelengths along the yellow to red range.

Spectral colors

- Colors that can be produced by visible light of a narrow band of wavelengths (monochromatic light) are called pure spectral colors.
- The spectrum is continuous, with no clear boundaries between one colour and the next
- The colours of the spectrum of white light are violet, indigo, blue, green, yellow, orange and red (VIBGYOR).



The white light is a mixture of different colours. Each colour is associated with light of a particular wavelength.

Red light has longer wavelengths than the blue light.

The angle of deviation by a prism is not the same for all the wavelength (colours) of light.

Hence the prism disperses white light into its constituent colours. The red is deviated least and the violet most.

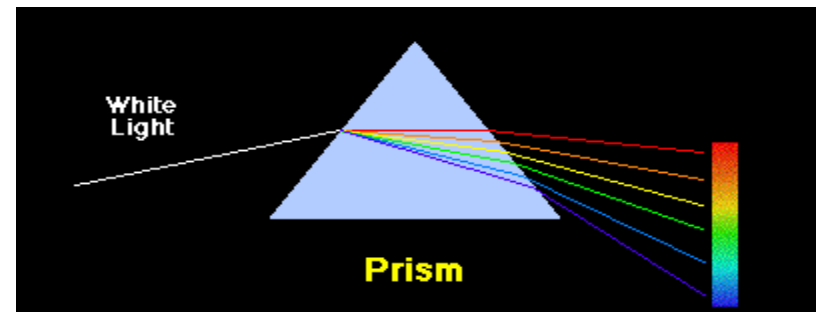
Color	Wavelength (nm)
Violet	400 – 440
Indigo	440 – 460
Blue	460 – 500
Green	500 – 570
Yellow	570 – 590
Orange	590 – 620
Red	620 – 720

Visible spectrum

- The **visible spectrum** is the portion of the electromagnetic spectrum that is visible to the human eye
- Electromagnetic radiation in this range of wavelengths is called **visible light** or simply light.
- A typical human eye will respond to wavelengths from about 390 to 700 nanometers
- In terms of frequency, this corresponds to a band in the vicinity of 430–770 THz.
- The spectrum does not contain all the colors that the human eyes and brain can distinguish.
- Unsaturated colors such as pink, or purple variations like magenta, for example, are absent because they can only be made from a mix of multiple wavelengths.
- Colors containing only one wavelength are also called pure colors or spectral colors.

Spectra

- When white light falls on a prism, placed in a spectrometer, the waves of different wavelengths are deviated to different directions by the prism.
- The image obtained in the field of view of the telescope consists of a number of coloured images of the slit.
- Such an image is called a spectrum.



A Quantitative Study



Sir Isaac Newton
1643 - 1727

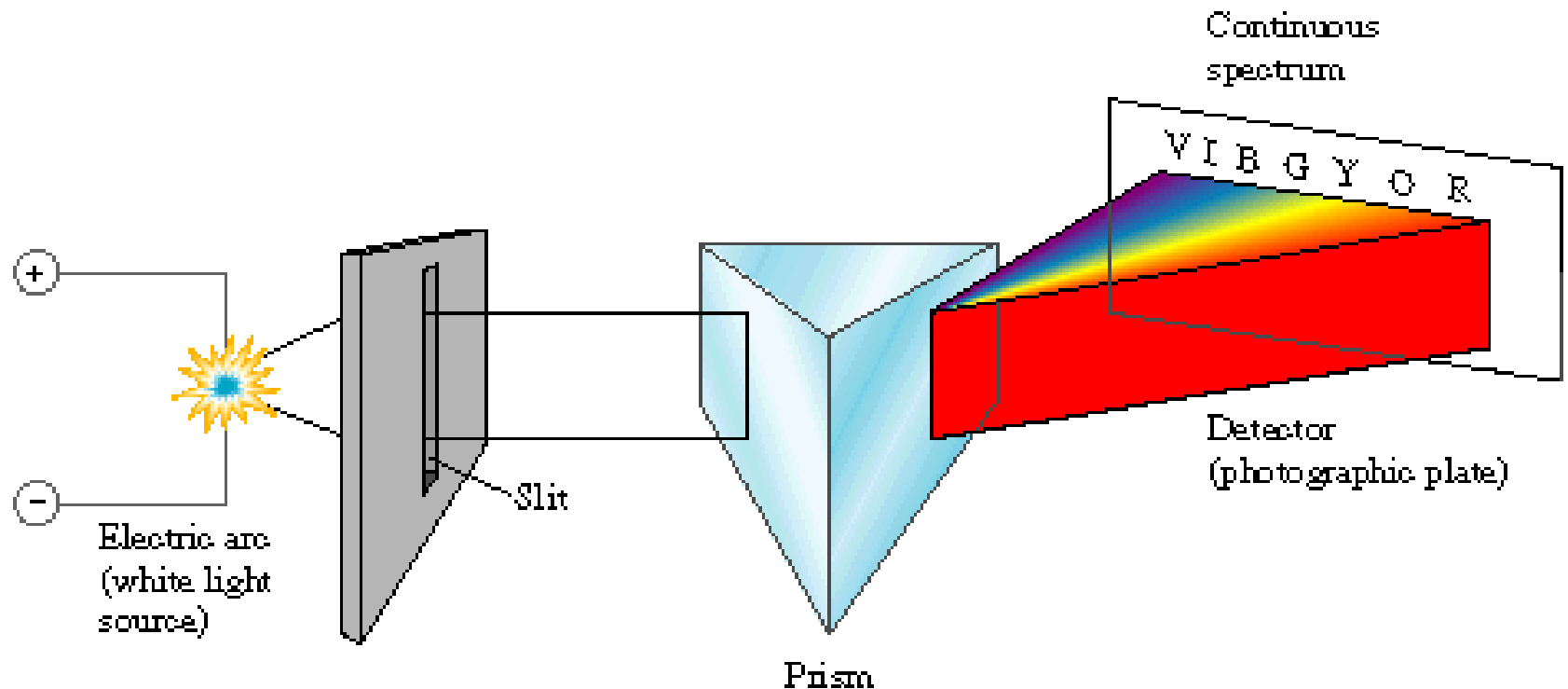
Sir Isaac Newton was one of the first people to study light scientifically.

In 1672, Newton directed a beam of white light through a triangular bar of glass, called a “prism”. He discovered that the light coming out of the prism was separated into bands of colors.

The arrangement of colors produced by a prism is called a “spectrum”.

Prior to this it was believed that “white light” was equal to purity.

Original Studies Of Light Used Only One Prism

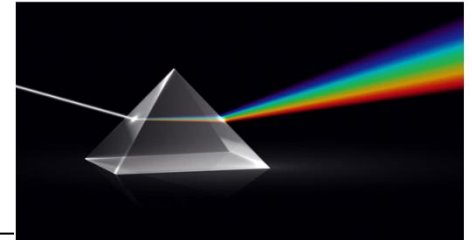


When a narrow band of light from a “white” light source is sent through a prism, a continuous spectrum containing all wavelengths of visible light is formed.

Sources of spectrum

- Sun is the biggest source of spectral light. The **sun** is the most obvious source of white light in our solar system.
- Not only visible light , it also radiates infrared and UV rays
- In visible region, it radiates all seven colors of VIBGYOR
- As for artificial sources, **fluorescent** light bulbs and white LEDs produce white light.
- Other light bulbs, like the incandescent lamp, do not produce white light.
- They produce light of much longer wavelengths along the yellow to red range.

Spectral colours



- Colours that can be produced by visible light of a narrow band of wavelengths (monochromatic light) are called **pure spectral colours**.
- The spectrum is continuous, with no clear boundaries between one colour and the next
- The colours of the spectrum of white light are **violet, indigo, blue, green, yellow, orange and red (VIBGYOR)**.

....Sources of spectrum

- Other commonly used sources of spectral light are
 - electric bulbs,
 - **sodium vapour lamp and**
 - **mercury vapour lamp,**
 - **bunsen flame and**
 - **carbon arc flame**

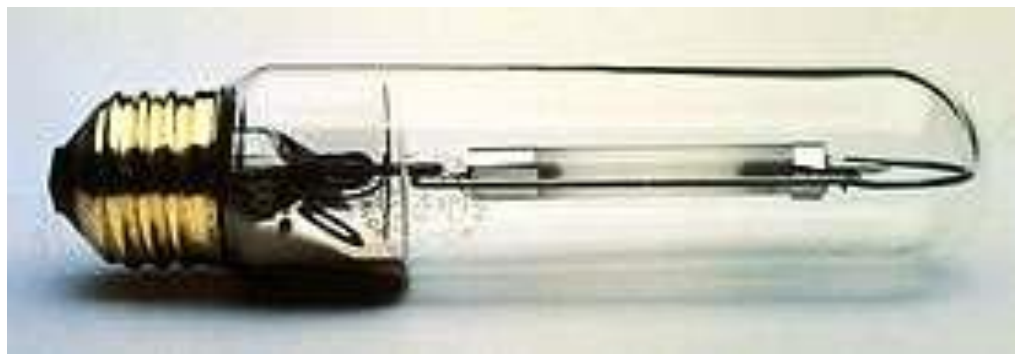
Sodium vapour lamp

- Principally the sodium vapour lamp consists of the bulb containing a small amount of **metallic sodium, neon gas, and two sets of electrodes connected in a pin type base.**
- The presence of neon gas serves to start the discharge, envelope is usually bent into U shape.
- The lamp operates at a temperature like 300°C



Mercury vapour lamps

- Mercury vapour lamps consist of **two electrodes**, that are **closely spaced**, placed in a **high-pressure gas medium**
- The gas and the electrodes are housed in an optically transparent, elliptically shaped enclosure **made of silica**
- The electrodes are made of **tungsten alloys**
- Tungsten alloys are chosen mainly due to their **high melting point**
- Along with **mercury**, there is a small amount of **argon or xenon that serves as a starter gas**.



**sodium vapour lamp and mercury vapour
lamp, bunsen flame and carbon arc falme**

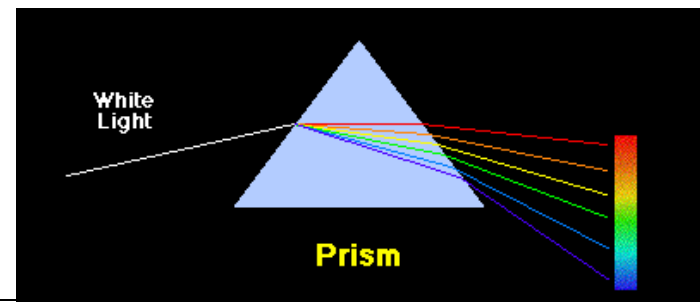
- Like sunlight, electric bulbs also radiate all seven colours of VIBGYOR but with different intensity
- A **sodium vapour lamp** is called **monochromatic** source of light as it provides light of yellow colour (characteristic wavelength $5890 \text{ \AA}$) only
- Actually other colours are also present in the spectrum of sodium lamp but are **too weak to be noticed**
- A **mercury vapour lamp** provides light of some of the colours of VIBGYOR but **not the complete spectrum**

Spectrum of sodium

- If the slit is illuminated with light from **sodium** vapour lamp, **two images of the slit** are obtained in the yellow region of the spectrum.
- These images are the emission lines of sodium having wave lengths $5896\text{\AA}$ and $5890\text{\AA}$.
- This is known as spectrum of sodium.

Types of spectra

- When white light falls on a prism, placed in a spectrometer, the waves of different wavelengths are deviated to different directions by the prism.
- The image obtained in the field of view of the telescope consists of a number of coloured images of the slit.
- Such an image is called a spectrum.
- The spectra obtained from different bodies can be classified into two types
 - emission spectra and
 - absorption spectra



What is Emission Spectrum?

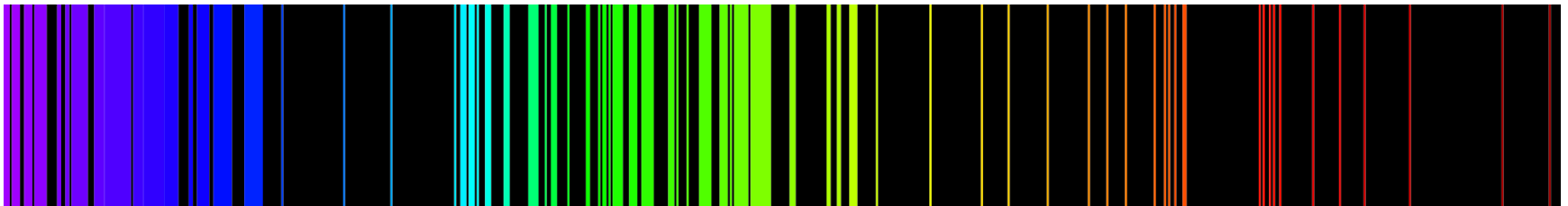
- When energy is absorbed by electrons of an atom, electrons move from lower energy levels to higher energy levels.
- These excited electrons have to radiate energy to return to ground states from the excited state, which is unstable.
- The emission spectrum is formed by the frequencies of these emitted light.

Emission Spectrum- explanation

- Whenever electromagnetic radiation interacts with atoms and molecules of matter, the electrons in these atoms may absorb energy and jump to a higher energy state, losing their stability.
- In order to regain their stability, they need to move from the higher energy state to the previous lower energy state.
- To accomplish this job, these atoms and molecules emit radiation in various regions of the electromagnetic spectrum.
- This spectrum of radiation emitted by electrons in the excited atoms or molecules is known as an ***emission spectrum***.

However, the ***emission spectrum*** of atoms in the gas phase, do not exhibit a continuous spread of wavelength from one colour to others

Rather, the emitted light consists of a **specific wavelength** having dark spaces existing between them. Such kind of spectra is known as atomic spectra or line spectra



Emission spectrum – Iron

What is Absorption Spectrum?

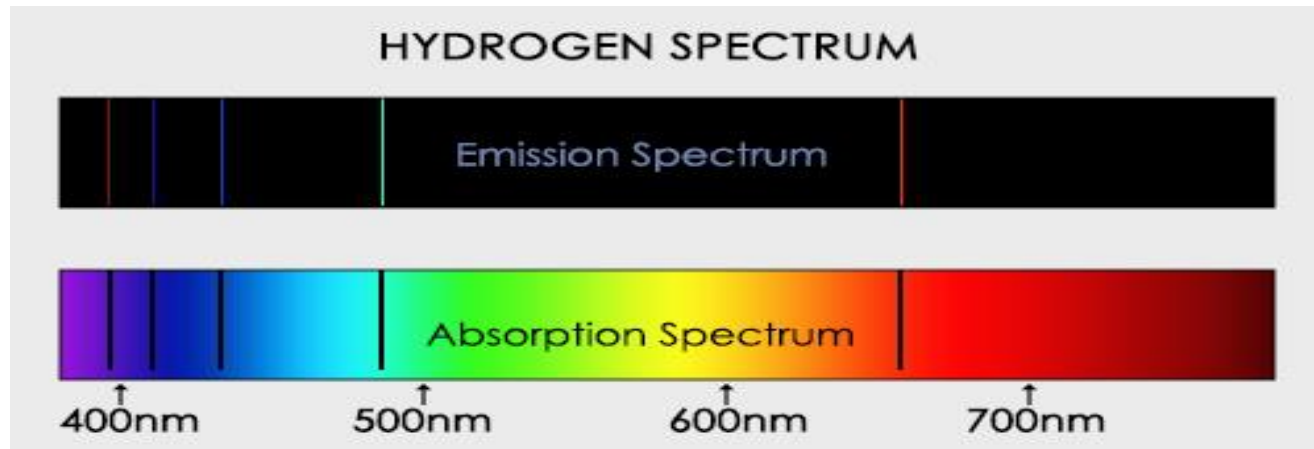
- An absorption spectrum is constituted by the frequencies of light transmitted with dark bands when energy is absorbed by the electrons in the ground state to reach higher energy states.

Absorption/ *continuous* Spectrum

- We observe that when a ray of white light falls on a prism it experiences refraction twice.
- Once when it travels from the rarer medium (air) to a denser medium (glass) and again from the denser medium (glass) to a rarer medium (air).
- Finally, we observe a band of colours, called spectrum, formed out of a ray of white light.
- If we observe this spectrum more closely, the colour having a smaller wavelength deviates the most and vice versa.
- Thus, a spectrum of colours ranging from red to violet is observed where **red having the longest wavelength suffers the least deviation.**
- This kind of spectrum is called a ***continuous spectrum*** as violet merges into blue, blue into green and so on.

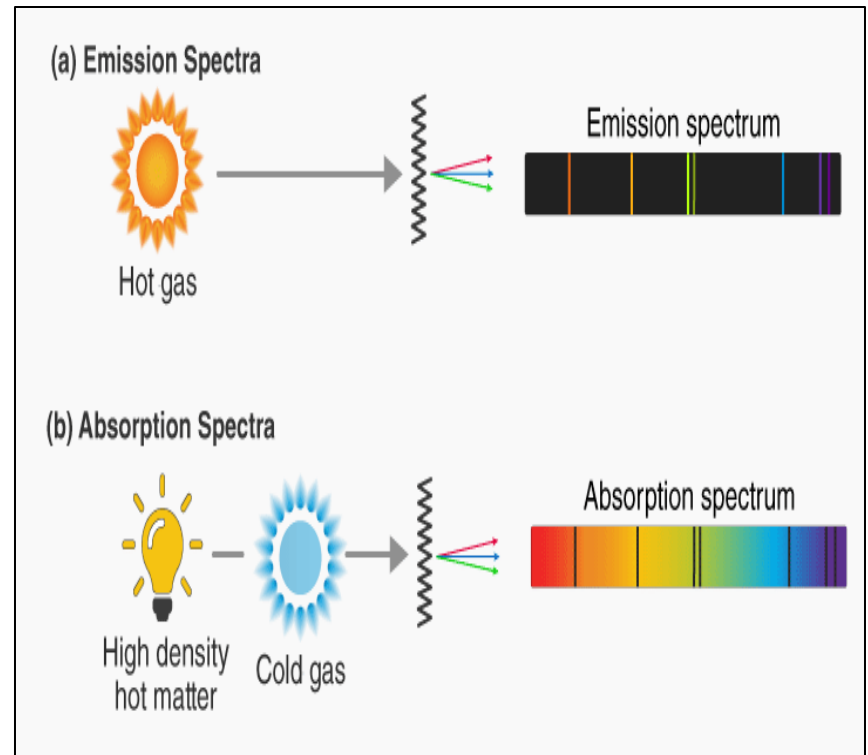
Emission Spectra VS Absorption Spectra

- The main difference between emission and absorption spectra is that an **emission** spectrum has **different coloured lines** in the spectrum, whereas an absorption spectrum has **dark lines** in the spectrum.



Emission Spectra VS Absorption Spectra

- Emission spectra can **emit all the colours** in an electromagnetic spectrum, while the absorption spectrum can have a **few colours missing** due to the redirection of absorbed photons



Emission Spectra

Produced when atoms release energy

Comprise coloured lines in the spectrum

It is helpful in figuring out the composition of a certain matter

The type of photons emitted is helpful in figuring out the kind of elements the substance is made of as each element radiates a different amount of energy and has a unique emission level

Absorption Spectra

Produced when atoms absorb energy

Comprise dark lines or gaps in the spectrum

Can be used to figure out the ability of certain objects to retain heat and its absorption level

The wavelengths of light absorbed is helpful in figuring out the number of substances in the sample.

ABSORPTION SPECTRA VERSUS EMISSION SPECTRA

An absorption spectrum can be defined as a spectrum obtained by transmitting electromagnetic radiation through a substance

Produced when atoms absorb energy

Show dark lines or gaps

An atom obtains a higher energy level when an absorption spectrum is given by that atom

Account for wavelengths absorbed by a substance

An emission spectrum can be defined as a spectrum of the electromagnetic radiation emitted by a substance

Produced when atoms release energy

Show colored lines

An emission spectrum is given when an excited atom obtains a lower energy level

Account for the wavelengths emitted by a substance

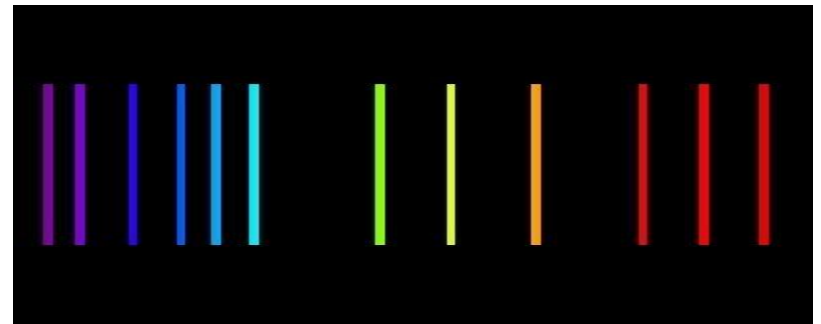
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Emission Spectrum & Absorption Spectrum

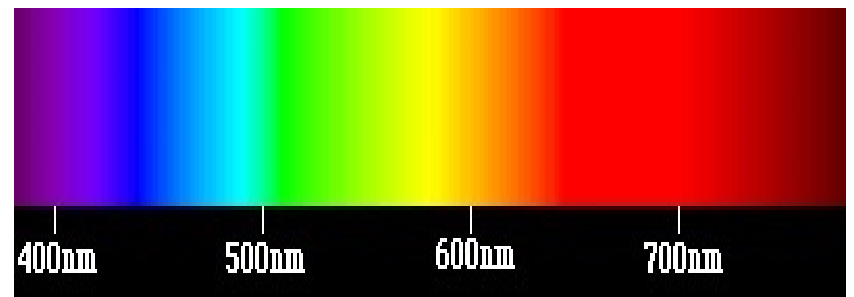
- An ***absorption spectrum*** is like a photographic negative of an emission spectrum.
- For observing the absorption spectrum, electromagnetic radiations are bombarded on a sample that absorbs radiation of certain wavelengths.
- The wavelength of radiation absorbed by the matter contributes to the missing wavelength which leaves dark spaces in the bright continuous spectrum.
- Each element has its unique line emission spectrum. The study of the emission spectrum or absorption spectrum is better known as spectroscopy.

Emission spectra

- When the **light emitted directly from a source** is examined with a **spectrometer**, the **emission spectrum** is obtained.
- Every source has its own characteristic emission spectrum.
- The emission spectrum is of three types.
 - Continuous spectrum
 - Line spectrum and
 - Band spectrum



Continuous spectrum



- It consists of unbroken luminous bands of all wavelengths containing all the colours from violet to red.
- These spectra **depend** only on the **temperature of the source** and is independent of the characteristic of the source.
- **Incandescent solids, liquids, Carbon arc, electric filament lamps** etc, give continuous spectra.

Line spectrum

- Line spectra are **sharp lines of definite wavelengths**.
- It is the characteristic of the **emitting substance**.
- It is used to **identify the gas**.
- Atoms in the gaseous state, i.e. free excited atoms **emit** line spectrum.
- The substance in atomic state such as **sodium in sodium vapour lamp, mercury in mercury vapour lamp** and gases in discharge tube give line spectra



Fig 5.4 Line spectrum of hydrogen

Band Spectrum

- It consists of a number of **bright bands with a sharp edge at one end but fading out at the other end.**
- Band spectra are obtained from **molecules.**
- It is the characteristic of the molecule.
- **Calcium or Barium salts in a bunsen flame and gases like carbon-di-oxide, ammonia and nitrogen in molecular state in the discharge tube give band spectra.**
- When the **bands are examined** with high resolving power spectrometer, each band is found to be made of a **large number of fine lines, very close to each other at the sharp edge but spaced out at the other end.**
- Using band spectra the **molecular structure of the substance** can be studied

Absorption Spectra

- When the light emitted from a source is **made to pass through an absorbing material and then examined with a spectrometer**, the obtained spectrum is called absorption spectrum.
- It is the characteristic of the **absorbing substance**.

Absorption Spectra

When the light emitted from a source is made to pass through an absorbing material and then examined with a spectrometer, the obtained spectrum is called absorption spectrum. It is the characteristic of the absorbing substance.

Absorption spectra is also of three types

1. continuous absorption spectrum
2. line absorption spectrum and
3. band absorption spectrum

Continuous absorption spectrum

A pure green glass plate when placed in the path of white light, absorbs everything except green and gives continuous absorption spectrum.

Line absorption spectrum

When light from the carbon arc is made to pass through sodium vapour and then examined by a spectrometer, a continuous spectrum of carbon arc with two dark lines in the yellow region is obtained

Band absorption spectrum

If white light is allowed to pass through iodine vapour or dilute solution of blood or chlorophyll or through certain solutions of organic and inorganic compounds, dark bands on continuous bright background are obtained. The band absorption spectra are used for making dyes.



Fig 5.5 Emission and absorption spectrum of sodium

Applications of emission and absorption spectra

- The study of spectra from stars and galaxies in astronomy is called *spectroscopy*.
- Spectroscopy is a tool widely used in astronomy to learn different things about astronomical objects.

Applications of emission and absorption spectra

To identify a substance:

- Every substance emits lights of different wavelengths.
- To identify the given substance, light is focused on it or the substance is heated.
- This causes the electrons to get excited and jump to a higher orbit.
- The energy emitted by these electrons while returning to their ground states is compared to the characteristic colors of the elements, and the chemical composition of the substance is determined.

Applications of emission and absorption spectra

To identify a substance and determine its concentration:

- An unknown substance can be identified by focusing light of a particular wavelength on it, and then studying the absorption spectrum of the substance.
- Since substances absorb light only from a particular wavelength or wavelength range, the wavelength of light focused on them is important.
- This spectrum can be compared with a set of reference values for identification.
- These reference values are known absorbance values of common elements and compounds.
- The concentration of the substance in the sample can also be determined.

Emission and Absorption Spectroscopy

1. Determining the atomic structure of a sample
2. Determining the metabolic structure of a muscle
3. Monitoring dissolved oxygen content in freshwater and marine ecosystems
4. Studying spectral emission lines of distant galaxies
5. Altering the structure of drugs to improve effectiveness.
6. Characterization of proteins.
7. Space exploration.
8. Respiratory gas analysis in hospitals.

UV spectrum

- Spectroscopy is the measurement and interpretation of electromagnetic radiation absorbed or emitted when the molecules or atoms or ions of a sample move from one energy state to another energy state
- UV spectroscopy is a type of absorption spectroscopy in which light of the ultra-violet region (200-400 nm) is absorbed by the molecule which results in the excitation of the electrons from the ground state to a higher energy state.

Principle of UV Spectroscopy

1. Basically, spectroscopy is related to the interaction of light with matter.
2. As light is absorbed by matter, the result is an increase in the energy content of the atoms or molecules.
3. When ultraviolet radiations are absorbed, this results in the excitation of the electrons from the ground state towards a higher energy state.
4. Molecules containing π -electrons or nonbonding electrons (n-electrons) can absorb energy in the form of ultraviolet light to excite these electrons to higher anti-bonding molecular orbitals.
5. The more easily excited the electrons, the longer the wavelength of light they can absorb. There are four possible types of transitions ($\pi-\pi^*$, $n-\pi^*$, $\sigma-\sigma^*$, and $n-\sigma^*$), and they can be ordered as follows: $\sigma-\sigma^* > n-\sigma^* > \pi-\pi^* > n-\pi^*$
6. The absorption of ultraviolet light by a chemical compound will produce a distinct spectrum that aids in the identification of the compound.

Applications of UV Spectroscopy

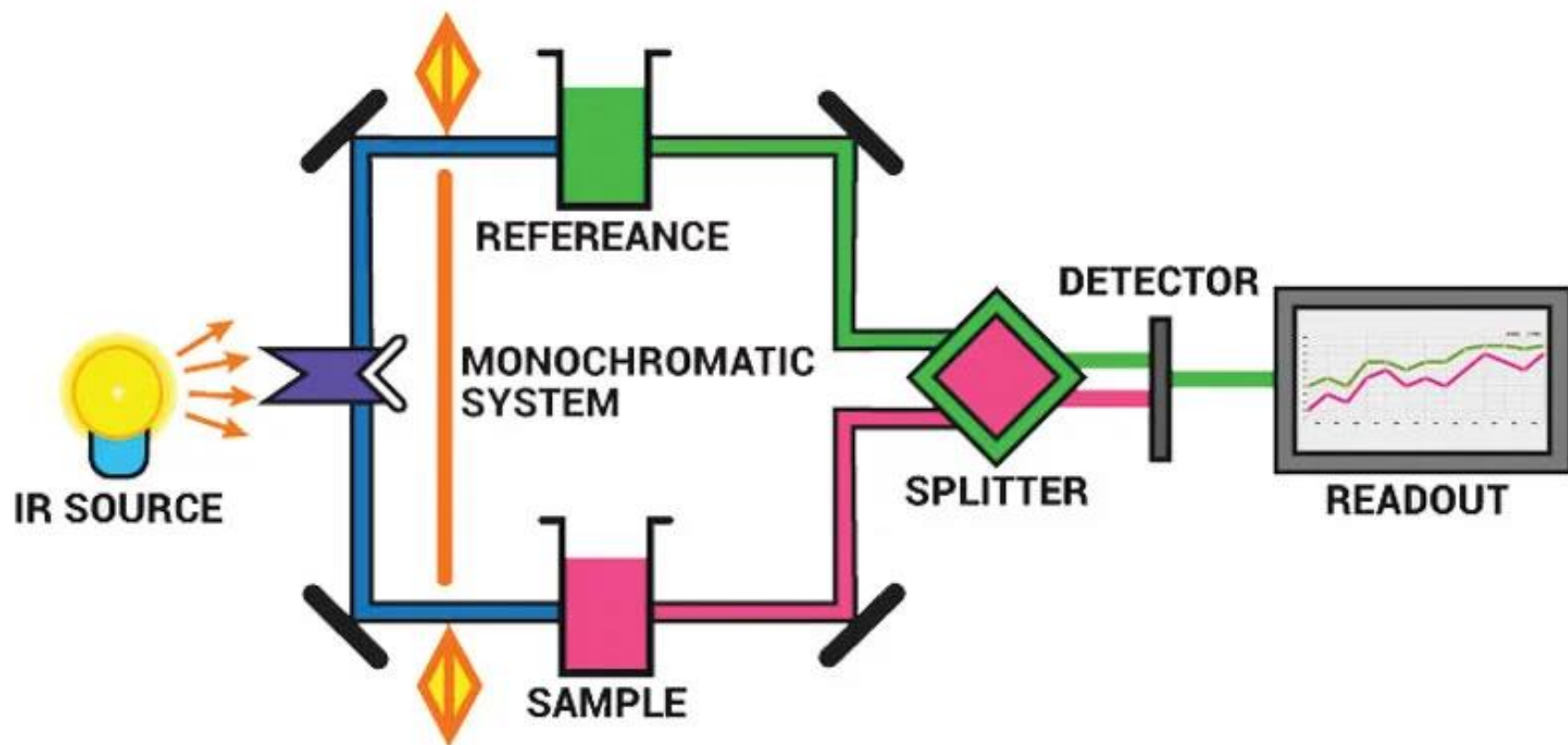
- **Detection of Impurities**
- It is one of the best methods for the determination of impurities in organic molecules.
- Additional peaks can be observed due to impurities in the sample and it can be compared with that of standard raw material.
- By also measuring the absorbance at a specific wavelength, the impurities can be detected.

Structure elucidation of organic compounds

- It is useful in the structure elucidation of organic molecules, such as in detecting the presence or absence of unsaturation, the presence of heteroatoms.
1. UV absorption spectroscopy can be used for the **quantitative determination of compounds** that absorb UV radiation.
 2. UV absorption spectroscopy can characterize those types of compounds that absorb UV radiation thus used in the qualitative determination of compounds. Identification is done by comparing the absorption spectrum with the spectra of known compounds.
 3. This technique is used to detect the presence or absence of a functional group in the compound. The absence of a band at a particular wavelength is regarded as evidence for the absence of particular group.
 4. Kinetics of reaction can also be studied using UV spectroscopy. The UV radiation is passed through the reaction cell and the absorbance changes can be observed.
 5. Many drugs are either in the form of raw material or in the form of the formulation. They can be assayed by making a suitable solution of the drug in a solvent and measuring the absorbance at a specific wavelength.

Infrared (IR) spectrum

- Infrared (IR) spectroscopy or vibrational spectroscopy is an analytical technique that takes advantage of the vibrational transitions of a molecule.
- It is one of the most common and widely used spectroscopic techniques employed mainly by **inorganic and organic chemists** due to its usefulness in **determining structures of compounds and identifying them**.
- The method or technique of infrared spectroscopy is conducted with an instrument called an infrared spectrometer (or **spectrophotometer**) to produce an infrared spectrum.



Principle of Infrared (IR) Spectroscopy

1. Infrared Spectroscopy is the analysis of infrared light interacting with a molecule.
2. The portion of the infrared region most useful for analysis of organic compounds have a wavelength range from 2,500 to 16,000 nm, with a corresponding frequency range from 1.9×10^{13} to 1.2×10^{14} Hz.
3. Photon energies associated with this part of the infrared (from 1 to 15 kcal/mole) are not large enough to excite electrons, but may induce vibrational excitation of covalently bonded atoms and groups.

4. Infrared spectrometers, similar in principle to other spectrometer, permit chemists to obtain absorption spectra of compounds that are a unique reflection of their molecular structure.
5. The fundamental measurement obtained in infrared spectroscopy is an infrared spectrum, which is a plot of measured infrared intensity versus wavelength (or frequency) of light.
6. IR Spectroscopy measures the vibrations of atoms, and based on this it is possible to determine the functional groups.
7. Generally, stronger bonds and light atoms will vibrate at a high stretching frequency (wavenumber).

Applications of Infrared (IR) Spectroscopy

- It has been of great significance to scientific researchers in many fields such as:
- Protein characterization
- Nanoscale semiconductor analysis and
- Space exploration.
- Analysis of gaseous, liquid or solid samples
- Identification of compounds
- Quantitative analysis
- Information regarding functional groups of molecules and constitution of molecules can be deduced from IR spectrum
- To know about interaction among molecules

The Solar Spectrum

- The Sun emits radiation from X-rays to radio waves, but the irradiance of solar radiation peaks in the visible wavelengths (see figure below).
- The units of irradiance are Joules per second per m^2 of surface that is illuminated per nm of wavelength (e.g., between 300nm and 301nm), or $\text{Watts m}^{-2} \text{ nm}^{-1}$ for this plot.
- These units are the units of spectral irradiance, which is also simply called irradiance, but as a function of wavelength.
- To get the total irradiance in units of watts m^{-2} , the spectral irradiance should be integrated over all the wavelengths.
- The sun radiates solar energy or sunlight by electromagnetic waves over a range of wavelengths known as the Solar Spectrum (290–2500 nanometres, where 1 nanometre = $1/1,000,000,000$ of a metre).

Solar Spectrum

- The sun radiates solar energy or sunlight by electromagnetic waves over a range of wavelengths known as the Solar Spectrum (290–2500 nanometres, where 1 nanometre = 1/1,000,000,000 of a metre).
- The solar spectrum is divided into three bands.
- These are
 - **Ultra-violet light (UV) – 290nm–380nm**
 - **Visible light – 380nm–780nm**
 - **Infra-red light – 780nm–2500nm**
- The energy distribution within the solar spectrum is approximately 2% UV, 47% visible and 51% infra-red.
- Only the visible light band is seen by the human eye

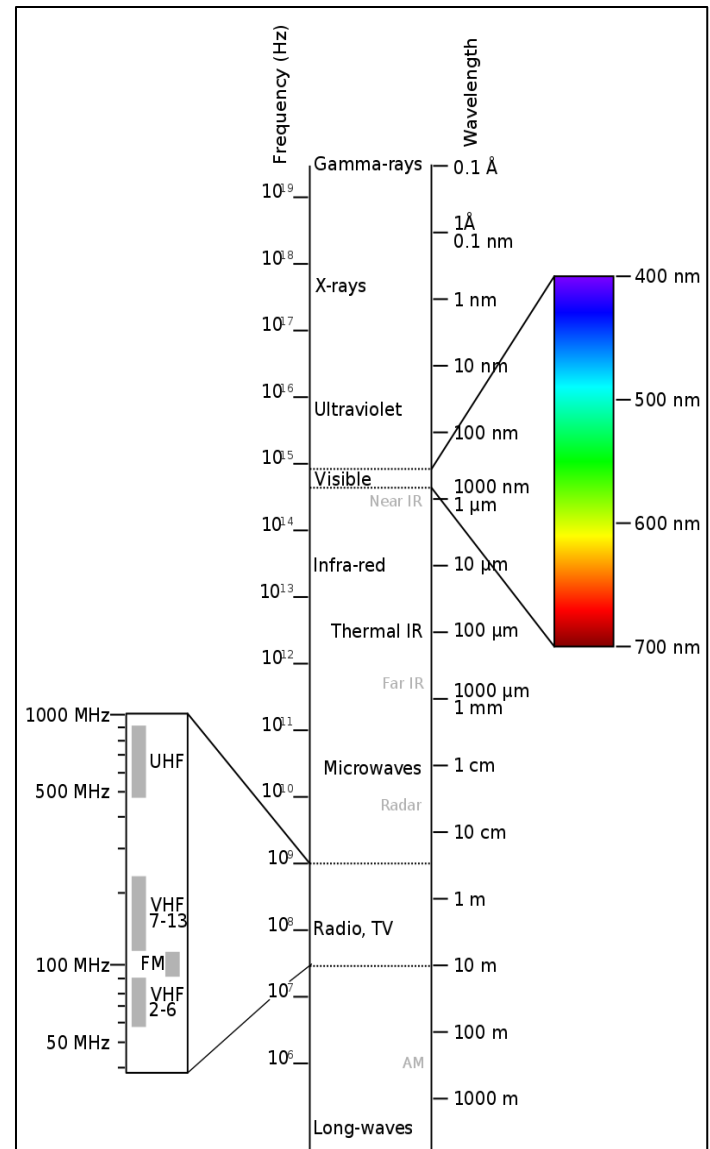
Fraunhofer lines

- If the solar spectrum is closely examined, it is found that it consists of **large number of dark lines**.
- These dark lines in the solar spectrum are called **Fraunhofer lines**. Solar spectrum is an example of **line absorption spectrum**.
- The **central core of the sun** is called photosphere which is at a very high temperature of the order of 14 million kelvin.
- It emits **continuous spectrum**. The sun's outer layer is called chromosphere.
- This is at a comparatively lower temperature at about 6000 K.
- It contains various elements in gaseous state.
- When light from the central core of the sun passes through sun's atmosphere, certain wavelengths are absorbed by the elements present in the chromosphere and the spectrum is marked by dark lines.
- By comparing the absorption spectra of various substances with the Fraunhofer lines in the solar spectrum, the elements present in the sun's atmosphere have been identified.

Electromagnetic spectrum

- Electromagnetic spectrum refers to **the full range of all frequencies of electromagnetic radiation**
- Devices used to measure an **electromagnetic spectrum are called spectrograph or spectrometer.**
- The **visible spectrum** is the part of the **electromagnetic spectrum** that can be seen by the human eye.
- Electromagnetic waves can be classified and **arranged according to their various wavelengths/frequencies;** this classification is known as the **electromagnetic spectrum.**

- For example, x-rays, gamma rays, infrared radiation and ultraviolet radiation are examples of electromagnetic waves.
- Only a small portion of the spectrum of wavelengths can be seen by the **human eye**.
- This visible portion of the electromagnetic spectrum is called the **visible spectrum**.
- This shows the full spectrum of electromagnetic radiation and highlights the small part of the spectrum that can be called the visible spectrum.



Region of the spectrum	Main interactions with matter
Radio	Collective oscillation of charge carriers in bulk material (plasma oscillation). An example would be the oscillatory travels of the electrons in an antenna .
Microwave through far infrared	Plasma oscillation, molecular rotation
Near infrared	Molecular vibration, plasma oscillation (in metals only)
Visible	Molecular electron excitation (including pigment molecules found in the human retina), plasma oscillations (in metals only)
Ultraviolet	Excitation of molecular and atomic valence electrons, including ejection of the electrons (photoelectric effect)
X-rays	Excitation and ejection of core atomic electrons, Compton scattering (for low atomic numbers)
Gamma rays	Energetic ejection of core electrons in heavy elements, Compton scattering (for all atomic numbers), excitation of atomic nuclei, including dissociation of nuclei
High-energy gamma rays	Creation of particle-antiparticle pairs . At very high energies a single photon can create a shower of high-energy particles and antiparticles upon interaction with matter.

- **Radio:** radio captures radio waves emitted by radio stations, bringing the favorite tunes. Radio waves are also emitted by stars and gases in space.
- **Microwave:** Microwave radiation will cook popcorn in just a few minutes, but is also used by astronomers to learn about the structure of nearby galaxies
- **Infrared:** Night vision goggles pick up the infrared light emitted by our skin and objects with heat. In space, infrared light helps to map the dust between stars.
- **Visible:** Our eyes detect visible light. Fireflies, light bulbs, and stars all emit visible light.
- **Ultraviolet:** Ultraviolet radiation is emitted by the Sun and are the reason skin tans and burns. "Hot" objects in space emit UV radiation as well.
- **X-ray:** A dentist uses X-rays to image teeth, and airport security uses them to see through the bag. Hot gases in the Universe also emit X-rays.
- **Gamma ray:** Doctors use gamma-ray imaging to see inside the body. The biggest gamma-ray generator of all is the Universe.

Measuring electromagnetic radiation

- Electromagnetic radiation can be expressed in terms of energy, wavelength, or frequency.
- Frequency is measured in cycles per second, or Hertz.
- Wavelength is measured in meters.
- $C = \gamma\lambda$
- Energy is measured in electron volts.
- $E = h\gamma$

Difference between Electromagnetic Radiation and Electromagnetic Spectrum

- EM radiation is an effect caused due to interactions between electric and magnetic fields.
- EM radiation is a qualitative concept, while EM spectrum is a quantitative measurement.
- EM spectrum is a quantitative method used to describe the EM radiation.
- The concept of EM radiation alone is useless. EM spectrum has many applications and usages.

Applications of Spectrum

- **Electromagnetic Spectrum**

1. **Gamma rays** – These rays are used to sanitize medical equipment and inhibit the growth of micro-organisms.
2. **X-rays** – These rays are used to visualize the inside of a body without making an incision. These rays are used for scanning purposes.
3. **Ultraviolet light** – These rays kill microbes, hence is used extensively to disinfect equipment.
4. **Visible light** – These rays help us to see things around us.
5. **Infrared rays** – As these rays can easily penetrate the skin, are used widely in cosmetic applications. It is used in remote controls, electrical heaters, and thermal cameras.
6. **Microwave** – It is widely used in microwave ovens to transmit the thermal energy required to cook food. Also used to guide airplanes.
7. **Radio waves** are used in radio and TV broadcasts.

Uses of electromagnetic spectrum

The following are some of the uses of electromagnetic waves.

1. Radio waves : These waves are used in radio and television communication systems. AM band is from 530 kHz to 1710 kHz. Higher frequencies upto 54 MHz are used for short waves bands.

Television waves range from 54 MHz to 890 MHz. FM band is from 88 MHz to 108 MHz. Cellular phones use radio waves in ultra high frequency (UHF) band.

2. Microwaves : Due to their short wavelengths, they are used in radar communication system. Microwave ovens are an interesting domestic application of these waves.

3. Infra red waves :

- i. Infrared lamps are used in physiotherapy.
- ii. Infrared photographs are used in weather forecasting.
- iii. As infrared radiations are not absorbed by air, thick fog, mist etc, they are used to take photograph of long distance objects.
- iv. Infra red absorption spectrum is used to study the molecular structure.

4. Visible light : Visible light emitted or reflected from objects around us provides information about the world. The wavelength range of visible light is 4000 Å to 8000 Å.

5. Ultra-violet radiations

i. They are used to destroy the bacteria and for sterilizing surgical instruments.

ii. These radiations are used in detection of forged documents, finger prints in forensic laboratories.

iii. They are used to preserve the food items.

iv. They help to find the structure of atoms.

6. X rays :

(i) X rays are used as a diagnostic tool in medicine.

(ii) It is used to study the crystal structure in solids.

7. γ -rays : Study of γ rays gives useful information about the nuclear structure and it is used for treatment of cancer.

Thank you